ANDI: Adaptive Norm-Distribution Interface

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Abstract

The optimization of deep neural networks is currently dominated by two paradigms: coordinate-wise adaptive methods (e.g., AdamW), which ignore parameter correlations, and higher-order structural methods (e.g., K-FAC, Muon), which enforce geometric constraints but suffer from super-linear computational complexity. We introduce the **Adaptive Norm-Distribution Interface (ANDI)**, a first-order optimizer that bridges this gap via structured preconditioning. ANDI applies an element-wise equilibration transformation derived from the additive equilibration of row and column norms, effectively approximating matrix balancing without iterative solvers or singular value decomposition. We prove that ANDI strictly maintains descent directions and provides an implicit trust region bounded by the gradient energy. Empirically, ANDI matches the convergence of spectral methods on ResNet-9 (CIFAR-10) while maintaining the $\mathcal{O}(N)$ computational profile of AdamW. Furthermore, on Transformer-based causal language modeling (NanoGPT), ANDI outperforms both diagonal and spectral baselines, suggesting that additive norm-equilibration serves as a superior inductive bias for attention-based architectures. Finally, we demonstrate scalability to the **8-billion parameter regime by fine-tuning Llama-3**, where ANDI exhibits rapid convergence within the constrained optimization subspaces of **Low-Rank Adaptation (LoRA)**. The code is available at `https://github.com/VladimerKhasia/ANDI`

1 Introduction

Stochastic Gradient Descent (SGD) and its adaptive variants, particularly Adam [8] and AdamW [9], serve as the backbone of modern deep learning. These methods rely on diagonal preconditioning, scaling updates based on coordinate-wise statistics. While computationally efficient ($\mathcal{O}(N)$), diagonal preconditioners treat the entries of weight matrices as independent variables, ignoring the rich structural correlations inherent in linear layers and attention heads.

Recent advances in structural optimization, such as Shampoo [5] and the spectral optimizer Muon [7], demonstrate that respecting the matrix geometry of gradients can significantly accelerate convergence. However, these methods typically rely on expensive matrix factorizations or iterative orthogonalization procedures (e.g., Newton-Schulz iterations), which introduce cubic complexity $\mathcal{O}(n^3)$ with respect to layer dimensions or substantial memory overheads.

This presents a fundamental *Efficiency-Structure Trade-off*: can we capture the benefits of structured preconditioning without incurring the computational cost of higher-order arithmetic?

Present Work. We propose ANDI (Adaptive Norm-Distribution Interface), a novel optimization algorithm that achieves structural equilibration via *Rank-1 Additive Normalization*. Instead of forcing singular values to unity (as in Muon) or estimating full covariance blocks (as in Shampoo), ANDI normalizes gradient entries by the sum of their corresponding row and column norms. This operation acts as a single-step additive approximation to Sinkhorn scaling [13], preventing individual neurons or feature channels from dominating the update signal.

Our contributions are as follows:

1. **Algorithm:** We formulate ANDI, a parameter-free preconditioning step that runs in $\mathcal{O}(N)$ time and $\mathcal{O}(1)$ auxiliary memory, making it strictly more efficient than spectral baselines.
2. **Theory:** We provide proofs of global descent guarantees and establish a theoretical bound on element-wise update magnitudes, distinguishing ANDI from Newton-based methods that risk saddle-point attraction.

3. **Empirical Validation:** We demonstrate that ANDI achieves Pareto-optimal performance across vision and language tasks, outperforming AdamW in convergence speed and exceeding the state-of-the-art version of Muon (D-Muon) in final loss on Transformer architectures.

2 Methodology

We introduce the Adaptive Norm-Distribution Interface (ANDI), a structured optimization algorithm designed to equilibrate the flow of gradients through high-dimensional weight matrices without destroying the sign structure of the descent direction. ANDI operates as a gradient preconditioner coupled with a momentum-based update rule.

2.1 The ANDI Preconditioner

Let $\mathcal{L}(\theta)$ denote the objective function with respect to parameters $\theta \in \mathbb{R}^d$. At time step t, let $\mathbf{G}_t = \nabla_\theta \mathcal{L}(\theta_t)$ be the stochastic gradient. For parameters representing linear transformations (e.g., Linear or Convolutional layers), we reshape the gradient into a matrix form $\mathbf{G}_t \in \mathbb{R}^{m\times n}$.

The ANDI preconditioner transforms $\mathbf{G}_t$ into a candidate update direction $\mathbf{U}_t$ via a two-stage process: *Additive Equilibration* and *Energy-Consistent Rescaling*.

Additive Equilibration. Standard matrix balancing techniques (e.g., Sinkhorn-Knopp) seek diagonal matrices D_1, D_2 such that $D_1\mathbf{G}D_2$ is doubly stochastic. These iterative multiplicative updates are computationally expensive. ANDI approximates this goal via a single-step additive normalization. We define the row norms $\mathbf{r} \in \mathbb{R}^m$ and column norms $\mathbf{c} \in \mathbb{R}^n$ as:

$$r_i = \|\mathbf{G}_{i,:}\|_2, \quad c_j = \|\mathbf{G}_{:,j}\|_2. \tag{1}$$

We compute the equilibrated gradient $\widehat{\mathbf{G}}$ element-wise:

$$\widehat{G}_{ij} = \frac{G_{ij}}{r_i + c_j + \epsilon}, \tag{2}$$

where ϵ is a small stability constant. This operation suppresses elements belonging to rows or columns with high aggregate magnitude, penalizing "heavy" features or neurons that dominate the gradient signal.

Energy-Consistent Rescaling. The equilibration step (2) alters the global magnitude of the update. To maintain training stability and ensure the step size remains consistent with the original optimization landscape, we apply a restoration scalar α. We define the target energy τ based on the original gradient norm:

$$\tau = \sqrt{\|\mathbf{G}\|_F^2 + 1}. \tag{3}$$

The scalar α is computed to map the equilibrated gradient magnitude to this target:

$$\alpha = \frac{\tau}{\|\widehat{\mathbf{G}}\|_F + \epsilon}. \tag{4}$$

The final preconditioned gradient is $\mathbf{H}_t = \alpha\widehat{\mathbf{G}}_t$.

2.2 Update Rule

ANDI is integrated with Nesterov momentum. Let $\mathbf{M}_t$ be the momentum buffer, η the learning rate, and μ the momentum coefficient. The update step is:

$$\mathbf{M}_{t+1} \leftarrow \mu\mathbf{M}_t + \mathbf{H}_t \tag{5}$$

$$\mathbf{V}_{t+1} \leftarrow \mathbf{H}_t + \mu\mathbf{M}_{t+1} \quad \text{(Nesterov correction)} \tag{6}$$

$$\theta_{t+1} \leftarrow \theta_t - \eta\mathbf{V}_{t+1} \tag{7}$$

3 Theoretical Analysis

In this section, we analyze the convergence properties and stability guarantees of ANDI. We define the transformation $\Phi : \mathbb{R}^{m\times n} \to \mathbb{R}^{m\times n}$ such that $\mathbf{H} = \Phi(\mathbf{G})$.

3.1 Descent Direction Guarantee

A critical requirement for any non-convex optimizer is that the update direction must maintain a positive correlation with the negative gradient. Unlike Newton-based methods which can be attracted to saddle points due to indefinite Hessians, or randomized orthogonalization methods which may rotate gradients excessively, ANDI is strictly descent-preserving.

Proposition 1 (Sign Preservation and Descent)**.** *For any non-zero gradient* $\mathbf{G} \in \mathbb{R}^{m\times n}$*, the preconditioned direction* $\mathbf{H}$ *satisfies* $\langle \mathbf{G}, \mathbf{H} \rangle > 0$*.*

Proof. Recall $\mathbf{H} = \alpha \widehat{\mathbf{G}}$. Expanding the inner product:

$$\langle \mathbf{G}, \mathbf{H} \rangle = \sum_{i,j} G_{ij} H_{ij} = \sum_{i,j} G_{ij} \left(\alpha \frac{G_{ij}}{r_i + c_j} \right). \tag{8}$$

Since $\alpha > 0$, norms $r_i, c_j \geq 0$, and assuming $\mathbf{G} \neq \mathbf{0}$ (implying denominators are positive):

$$\langle \mathbf{G}, \mathbf{H} \rangle = \alpha \sum_{i,j} \frac{G_{ij}^2}{r_i + c_j}. \tag{9}$$

Since $G_{ij}^2 \geq 0$ for all i, j and strictly positive for at least one entry, the sum is strictly positive. Thus, $\mathbf{H}$ is a valid descent direction. □

3.2 Implicit Trust Region and Stability

Standard adaptive methods (like Adam) bound updates by ± 1 (sign) or scale by moving averages. ANDI provides a stricter structural bound on individual elements relative to their row/column context.

Proposition 2 (Element-wise Boundedness)**.** *Let* $\widehat{\mathbf{G}}$ *be the equilibrated matrix defined in Eq. (2). For any entry* (i, j)*,* $|\widehat{G}_{ij}| \leq 1$*. Further, if* $m, n > 1$*,* $|\widehat{G}_{ij}| < 1$ *for generic non-sparse gradients.*

Proof. By definition, $r_i = \sqrt{\sum_k G_{ik}^2} \geq |G_{ij}|$ and $c_j = \sqrt{\sum_k G_{kj}^2} \geq |G_{ij}|$.

$$|\widehat{G}_{ij}| = \frac{|G_{ij}|}{r_i + c_j} \leq \frac{|G_{ij}|}{|G_{ij}| + |G_{ij}|} = \frac{1}{2}. \tag{10}$$

Even in the worst case where a row and column contain only zeros except for the intersection G_{ij}, the denominator is $2|G_{ij}|$, bounding the value at 0.5. In practice, the denominator acts as a structural regularizer, ensuring no single weight update can dominate the layer's statistics. □

3.3 Adaptive Regime Analysis

The scalar α introduces a non-linear scaling dependent on the gradient magnitude $\|\mathbf{G}\|_F$. We analyze two limiting regimes:

Case 1: Vanishing Gradients ($\|\mathbf{G}\|_F \to 0$). As $\|\mathbf{G}\|_F \to 0$, $\tau \to 1$. Consequently, the update $\mathbf{H} \approx \frac{1}{\|\widehat{\mathbf{G}}\|_F} \widehat{\mathbf{G}}$. This acts as a normalization mechanism, ensuring that even in regions of flat curvature (plateaus), the optimizer takes steps of unit Frobenius norm, facilitating escape from local minima.

Case 2: Exploding Gradients ($\|\mathbf{G}\|_F \to \infty$). As $\|\mathbf{G}\|_F \to \infty$, $\tau \approx \|\mathbf{G}\|_F$. Assuming the structure of $\mathbf{G}$ is stable, $\|\widehat{\mathbf{G}}\|_F$ converges to a structural constant K. Thus, $\mathbf{H} \approx \frac{\|\mathbf{G}\|_F}{K} \widehat{\mathbf{G}}$. Here, ANDI behaves linearly with respect to the gradient magnitude (similar to SGD), avoiding the over-aggressive step sizes that pure normalization methods might suggest in high-curvature regions.

Remark 1. *ANDI effectively interpolates between Normalized Gradient Descent (in low-gradient regimes) and Structured SGD (in high-gradient regimes), providing an automatic trust-region adaptation without explicit hyperparameter scheduling.*

3.4 Homogeneous Embedding and Non-Vanishing Manifold Velocity

The energy-consistent restoration scalar $\tau = \sqrt{\|\mathbf{G}\|_F^2 + 1}$ corresponds to a projection from the Euclidean gradient space onto a unit-hyperboloid manifold $\mathcal{H}^n$ embedded in $\mathbb{R}^{mn+1}$. Let $\tilde{\mathbf{G}} = [\text{vec}(\mathbf{G}); 1] \in \mathbb{R}^{mn+1}$ be an augmented representation.

The scaling by τ ensures that the optimizer maintains a non-zero lower bound on the update magnitude, preventing the "stagnation" typical of first-order methods in quasi-flat regions (plateaus).

Proposition 3 (Lower Bound on Update Energy). *For any $\mathbf{G} \in \mathbb{R}^{m\times n}$, the norm of the preconditioned update $\mathbf{H}$ satisfies:*

$$\|\mathbf{H}\|_F \geq \frac{\sqrt{\|\mathbf{G}\|_F^2 + 1}}{\|\widehat{\mathbf{G}}\|_F} > 0. \tag{11}$$

As $\|\mathbf{G}\|_F \to 0$, the update norm $\|\mathbf{H}\|_F$ converges to $\|\widehat{\mathbf{G}}\|_F^{-1}$, which is strictly positive for non-degenerate $\mathbf{G}$.

This property implies that ANDI effectively operates as *Normalized Gradient Descent* (NGD) in the vanishing gradient limit. Unlike standard NGD, which is discontinuous at the origin, the ANDI operator $\Phi(\mathbf{G})$ is a smooth map, as the $+1$ term acts as a C^∞ regularizer for the norm function.

3.5 Spectral Conditioning via Rank-1 Marginal Balancing

The choice of the additive denominator $D_{ij} = r_i + c_j$ is theoretically grounded in the minimization of the condition number of the weight update. It is a known result in numerical linear algebra [14] that diagonal scaling $\mathbf{D}_1\mathbf{A}\mathbf{D}_2$ which equilibrates row and column norms minimizes the condition number κ within a factor of $\sqrt{n}$ of the optimum.

ANDI's additive equilibration acts as a first-order approximation to the Sinkhorn-Knopp algorithm [13]. By the AM-GM inequality, $r_i + c_j \geq 2\sqrt{r_i c_j}$, establishing that ANDI's additive scaling is a strictly conservative majorant of the geometric mean scaling used in symmetric matrix balancing.

Proposition 4 (Spectral Homogenization). *Let $\sigma_{\max}(\mathbf{H})$ be the spectral radius of the update. The operator Φ acts as a contraction on the outlier singular values of $\mathbf{G}$ by penalizing entries associated with high-energy marginals r_i, c_j. This effectively homogenizes the marginal distribution of the gradient energy, acting as a proxy for second-order whitening without the $\mathcal{O}(n^3)$ cost of Eigendecomposition or Matrix Inversion.*

3.6 Smoothness and Lipschitz Continuity of the Operator

To guarantee convergence in stochastic settings, the preconditioning operator must be Lipschitz continuous. We define the ANDI transformation as $\Phi : \mathbb{R}^{m\times n} \to \mathbb{R}^{m\times n}$.

Proposition 5 (Lipschitz Continuity). *The operator $\Phi(\mathbf{G})$ is Lipschitz continuous on any compact set $K \subset \mathbb{R}^{m\times n}$ where $\|\mathbf{G}\|_F > 0$.*

Proof. The map $\Phi(\mathbf{G}) = \alpha(\mathbf{G}) \cdot (\mathbf{G} \oslash \mathbf{D}(\mathbf{G}))$ is a composition of differentiable functions. The denominator $D_{ij} = r_i + c_j + \epsilon$ is bounded away from zero by ϵ. The scalar $\alpha(\mathbf{G})$ involves the Frobenius norm, which is differentiable everywhere except at the origin; however, the term $\sqrt{\|\mathbf{G}\|_F^2 + 1}$ is differentiable for all $\mathbf{G}$. Since the partial derivatives of Φ are bounded on K, the operator is L-Lipschitz. $\square$

This result is fundamental for stability: it ensures that small perturbations in the stochastic gradient (noise) do not lead to disproportionately large jumps in the preconditioned update, a common failure mode in strictly interpolating or higher-order optimizers.

4 Algorithm Specification

We formally present the proposed method in Algorithm 1. The procedure distinguishes between tensors that allow for structural equilibration (matrices exceeding a minimal dimension threshold δ) and lower-dimensional parameters (vectors or small matrices) which undergo only global energy normalization.

Algorithm 1: Adaptive Norm-Distribution Interface (ANDI)

Input: Parameters θ_0, Learning rate η, Momentum μ, Weight Decay λ, Dimension Threshold δ, Stability $\epsilon = 10^{-8}$.
Initialize: Momentum buffer $\mathbf{M}_0 \leftarrow \mathbf{0}$.
for $t = 1 \dots T$ **do**
  Compute Gradients: $\mathbf{G}_t \leftarrow \nabla_\theta \mathcal{L}(\theta_{t-1})$;
  for *each parameter tensor* $\mathbf{g} \in \mathbf{G}_t$ **do**
    `// Step 0:  Decoupled Weight Decay`
    $\theta[\mathbf{g}] \leftarrow (1 - \lambda\eta) \cdot \theta[\mathbf{g}]$;
    `// Compute norms and target energy`
    $\nu \leftarrow \|\mathbf{g}\|_F$;
    $\tau \leftarrow \sqrt{\nu^2 + 1}$;
    Let $(d_1, \dots, d_k)$ be dimensions of $\mathbf{g}$.
    if $k > 1$ **then**
      `// Matricization:  Flatten dims` $2 \dots k$
      $\mathbf{G}_{mat} \leftarrow \text{Reshape}(\mathbf{g}, (d_1, \prod_{i=2}^{k} d_i))$;
      $(m, n) \leftarrow \text{shape}(\mathbf{G}_{mat})$;
      `// Check dimensions against threshold`
      if $m \geq \delta$ ***and*** $n \geq \delta$ **then**
        `// Structural Equilibration`
        $\mathbf{r} \leftarrow \|\mathbf{G}_{mat}^{(i,:)}\|_2 + \epsilon$;
        $\mathbf{c} \leftarrow \|\mathbf{G}_{mat}^{(:,j)}\|_2 + \epsilon$;
        `// Broadcasting Addition`
        $\mathbf{D}_{ij} \leftarrow r_i + c_j$;
        $\widehat{\mathbf{g}} \leftarrow \mathbf{G}_{mat} \oslash \mathbf{D}$;
        `// Rescale and Restore Shape`
        $\alpha \leftarrow \tau / (\|\widehat{\mathbf{g}}\|_F + \epsilon)$;
        $\mathbf{g}_{final} \leftarrow (\alpha \cdot \widehat{\mathbf{g}}).\text{view}(d_1, \dots, d_k)$;
      else
        $\mathbf{g}_{final} \leftarrow \mathbf{g} \cdot (\tau / (\nu + \epsilon))$;
      end
    else
      `// Global Normalization (Vectors)`
      $\mathbf{g}_{final} \leftarrow \mathbf{g} \cdot (\tau / (\nu + \epsilon))$;
    end
    `// Nesterov Momentum Update`
    $\mathbf{m}_{buf} \leftarrow \mathbf{M}_{t-1}[\mathbf{g}]$;
    $\mathbf{m}_{new} \leftarrow \mu \cdot \mathbf{m}_{buf} + \mathbf{g}_{final}$;
    if *Nesterov* **then**
      $\mathbf{u} \leftarrow \mathbf{g}_{final} + \mu \cdot \mathbf{m}_{new}$;
    else
      $\mathbf{u} \leftarrow \mathbf{m}_{new}$;
    end
    `// Parameter Step`
    $\theta_t[\mathbf{g}] \leftarrow \theta_{t-1}[\mathbf{g}] - \eta \cdot \mathbf{u}$;
    $\mathbf{M}_t[\mathbf{g}] \leftarrow \mathbf{m}_{new}$;
  end
end

5 Computational Complexity and Guarantees

We analyze the asymptotic time complexity of ANDI per optimization step relative to standard baselines (AdamW) and spectral methods (Muon). Let $W \in \mathbb{R}^{m \times n}$ denote a weight matrix parameter. Let $N = m \times n$ be the total number of elements.

5.1 ANDI Complexity

The computational cost of ANDI is dominated by the calculation of row and column norms and the element-wise division.

1. **Norm Reduction:** Calculating $\mathbf{r}$ requires summing squares along rows, a reduction operation of $\mathcal{O}(N)$. Similarly, calculating $\mathbf{c}$ is $\mathcal{O}(N)$.
2. **Broadcasting and Division:** Computing $D_{ij} = r_i + c_j$ and the subsequent division G_{ij}/D_{ij} involves $\mathcal{O}(1)$ operations per element, totaling $\mathcal{O}(N)$.
3. **Global Scaling:** Frobenius norms and scalar multiplications are $\mathcal{O}(N)$.

Thus, the total time complexity per step is:

$$T_{\text{ANDI}} \in \mathcal{O}(N) = \mathcal{O}(mn). \tag{12}$$

Crucially, ANDI relies only on element-wise and reduction operations, making it **memory-bandwidth bound** rather than compute bound, similar to SGD.

5.2 Baseline Comparisons

AdamW. AdamW maintains two moment estimates (mean and variance). It performs element-wise squaring, square roots, and division.

$$T_{\text{AdamW}} \in \mathcal{O}(N). \tag{13}$$

ANDI shares the same linear asymptotic complexity class as AdamW. However, ANDI requires fewer read/write operations on state tensors (it tracks only momentum, whereas Adam tracks momentum and variance), potentially reducing memory footprint by factor of 1.5× relative to Adam.

Muon (Newton-Schulz). Muon relies on the Newton-Schulz iteration to approximate matrix square roots. For a matrix $G \in \mathbb{R}^{m\times n}$ (assuming $m \leq n$), the dominant operation is Matrix-Matrix Multiplication (MMM). The standard Newton-Schulz iteration involves computing $A = XX^T$ and BX, which are operations of order $\mathcal{O}(m^2 n)$. If K is the number of Newton-Schulz steps (typically $K = 5$):

$$T_{\text{Muon}} \in \mathcal{O}(K \cdot m^2 n). \tag{14}$$

In the case of square matrices where $m \approx n$, Muon exhibits cubic complexity $\mathcal{O}(n^3)$, whereas ANDI remains quadratic $\mathcal{O}(n^2)$ (linear in the number of parameters).

5.3 Summary of Guarantees

Table 1 summarizes the theoretical properties. ANDI occupies a distinct pareto-optimal point: it offers structured preconditioning (utilizing the matrix geometry) while maintaining the $\mathcal{O}(N)$ cost profile of coordinate-wise methods.

Table 1: Comparison of Optimization Algorithms per Step

Algorithm	**Complexity**	**Preconditioning**	**State Memory**
SGD	$\mathcal{O}(N)$	None	1× (Momentum)
AdamW	$\mathcal{O}(N)$	Diagonal (Coordinate-wise)	2× (Mom + Var)
Muon	$\mathcal{O}(Km^2n)$	Spectral (SVD-approx)	1× (Momentum)
ANDI (Ours)	$\mathcal{O}(\mathbf{N})$	**Rank-1 Additive**	**1× (Momentum)**

5.4 Theoretical Analysis: Additive Energy Equilibration

The empirical efficacy of ANDI stems from its ability to approximate the geometric benefits of matrix balancing and spectral preconditioning without incurring their $\mathcal{O}(n^3)$ computational costs. In this section, we analyze the algebraic properties of the equilibration operator $\Phi : \mathbb{R}^{m\times n} \to \mathbb{R}^{m\times n}$, defined element-wise as:

$$\Phi(\mathbf{G})_{ij} = \frac{G_{ij}}{\|\mathbf{G}_{i,:}\|_2 + \|\mathbf{G}_{:,j}\|_2 + \epsilon}. \tag{15}$$

We establish that this operator acts as a conservative structural regularizer that mitigates the hubness phenomenon and homogenizes the spectral landscape.

5.4.1 Majorization of Geometric Balancing

Standard matrix equilibration techniques, such as the Ruiz algorithm [11] or Sinkhorn-Knopp [?], typically seek to normalize a matrix $\mathbf{G}$ via diagonal scaling matrices $\mathbf{D}_1, \mathbf{D}_2$. A single step of symmetric geometric balancing transforms an entry G_{ij} by the geometric mean of its marginal norms: $G_{ij}^{\text{geo}} = G_{ij}(\|\mathbf{G}_{i,:}\|_2\|\mathbf{G}_{:,j}\|_2)^{-1/2}$.

ANDI employs an additive denominator. We show that this formulation imposes a strictly conservative trust region relative to geometric scaling.

Proposition 6 (Conservative Regularization Bound)**.** *Let $r_i = \|\mathbf{G}_{i,:}\|_2$ and $c_j = \|\mathbf{G}_{:,j}\|_2$. For any entry with non-zero marginals, the update magnitude of ANDI is bounded by half the magnitude of the geometric equilibration step.*

Proof. By the Inequality of Arithmetic and Geometric Means (AM-GM), for real non-negative r_i, c_j:

$$\frac{r_i + c_j}{2} \geq \sqrt{r_i c_j}. \tag{16}$$

Equality holds if and only if $r_i = c_j$. Taking the reciprocal reverses the inequality:

$$\frac{1}{r_i + c_j} \leq \frac{1}{2\sqrt{r_i c_j}}. \tag{17}$$

Multiplying both sides by the gradient magnitude $|G_{ij}|$ yields:

$$|\Phi(\mathbf{G})_{ij}| \leq \frac{1}{2}\left|\frac{G_{ij}}{\sqrt{r_i c_j}}\right| = \frac{1}{2}|G_{ij}^{\text{geo}}|. \tag{18}$$

□

Remark 2. ***Implication for Non-Convex Optimization:*** *The factor $\frac{1}{2}$ acts as a geometric dampening coefficient on the local curvature. Crucially, this constant scaling is absorbed by the energy restoration scalar α (Eq. 4) and does not suppress the global effective learning rate. Instead, it modifies the relative shape of the update: ANDI enforces a tighter local trust region specifically when row and column norms are imbalanced ($r_i \neq c_j$), favoring stability over the aggressive orthogonalization typical of spectral methods.*

5.4.2 Asymptotic Suppression of Gradient Hubs

High-dimensional optimization is frequently degraded by the *Hubness Phenomenon* [10], where specific neurons (rows) or attention heads accumulate disproportionately high gradient norms. We prove that ANDI asymptotically saturates the update norm for such outliers, effectively acting as a soft-clipping mechanism.

Proposition 7 (Asymptotic Row-Norm Saturation)**.** *Consider a gradient matrix $\mathbf{G}$ where a specific row k dominates the local geometry such that $r_k \to \infty$ while column norms c_j remain bounded by some constant C. The Euclidean norm of the updated row vector $\widehat{\mathbf{g}}_k = \Phi(\mathbf{G})_{k,:}$ converges to unity from below.*

Proof. The j-th entry of the updated row is $\widehat{G}_{kj} = \frac{G_{kj}}{r_k + c_j}$. The squared Euclidean norm of the updated row is:

$$\|\widehat{\mathbf{g}}_k\|_2^2 = \sum_{j=1}^{n}\left(\frac{G_{kj}}{r_k + c_j}\right)^2. \tag{19}$$

In the limit $r_k \to \infty$, the term c_j in the denominator becomes negligible relative to r_k:

$$\lim_{r_k\to\infty}\|\widehat{\mathbf{g}}_k\|_2^2 = \lim_{r_k\to\infty}\sum_{j=1}^{n}\frac{G_{kj}^2}{r_k^2} = \frac{1}{r_k^2}\sum_{j=1}^{n}G_{kj}^2. \tag{20}$$

Substituting the definition $r_k^2 = \sum_{j=1}^{n} G_{kj}^2$, we obtain:

$$\lim_{r_k \to \infty} \|\widehat{\mathbf{g}}_k\|_2^2 = \frac{r_k^2}{r_k^2} = 1. \tag{21}$$

Since $c_j \geq 0$, the denominator $r_k + c_j \geq r_k$, implying $\|\widehat{\mathbf{g}}_k\|_2 \leq 1$ for all finite r_k, c_j. □

Remark 3. *This property ensures that no single feature dimension can dominate the update step, strictly preventing the "exploding gradient" problem at the granularity of individual rows/columns. This provides a theoretical justification for ANDI's superior performance on Transformer architectures (Sec. 6), which are sensitive to scale variations in residual streams.*

5.4.3 Conditioning via Marginal Homogenization

While ANDI does not explicitly enforce orthogonality, it acts as a computationally efficient proxy for spectral preconditioning by homogenizing the marginal distributions of the gradient matrix.

Van der Sluis [14] established that for any matrix $\mathbf{A}$, minimizing the condition number $\kappa(\mathbf{D}_1\mathbf{A}\mathbf{D}_2)$ is closely related to balancing the row and column norms. Specifically, scaling a matrix such that all row and column norms are equal brings the condition number within a factor of $\sqrt{n}$ of the optimal diagonal scaling.

ANDI penalizes entries G_{ij} where the aggregate marginal $r_i + c_j$ is large. This operation inherently reduces the variance between the effective row and column norms of the update $\widehat{\mathbf{G}}$. By driving the optimization trajectory through a sequence of updates with homogenized marginals, ANDI implicitly smooths the spectral condition number of the effective step, facilitating faster convergence along high-curvature directions without the $\mathcal{O}(n^3)$ cost of Singular Value Decomposition.

6 Experiments

To empirically validate the theoretical properties of ANDI, we evaluate the optimizer across four distinct modalities: non-convex dimensionality reduction (AUTOENCODER), computer vision (RESNET), causal language modeling (GPT), and large-scale instruction fine-tuning (LLAMA).

6.1 Experimental Setup

Following the rigorous benchmarking protocols established by Semenov et al. [12], we compare ANDI against two primary baselines:

1. **AdamW** [9]: The standard diagonal preconditioner. We utilize a Cosine learning rate schedule, which is the community standard for coordinate-wise methods.
2. **Dispersed Muon (D-Muon)** [4, 12]: A state-of-the-art spectral optimizer. Unlike the original Muon, D-Muon applies Newton-Schulz orthogonalization only to 2D tensors while using an AdamW backend for vectors, alongside decoupled weight decay. We utilize the Warmup-Stable-Decay (WSD) scheduler, as Muon-based methods have been shown to underperform with standard cosine decay [12].

Methodology. Experiments were conducted on a single NVIDIA GPU. We report **Validation Loss** as the primary metric to assess generalization. To quantify computational overhead, we report Wall-Clock Time. Hyperparameters were tuned via grid search for ANDI and both baselines. Detailed configurations are provided in Appendix A.

6.2 Task 1: Causal Language Modeling (GPT)

We train a NanoGPT transformer [15] on the TinyShakespeare corpus.

Results. As summarized in Table 2, ANDI ($\eta = 0.09$) achieves the best performance, reaching a validation loss of **1.535**, strictly outperforming D-Muon (1.555) and AdamW (1.577). Computationally, ANDI remains efficient. The total wall-clock time for ANDI (108s) is comparable to AdamW (103s), whereas D-Muon incurs a $\sim 23\%$ overhead (127s) due to the $\mathcal{O}(N^3)$ cost of Newton-Schulz iterations. This suggests ANDI's row-column normalization approximates the geometric benefits of spectral methods for Attention layers without the cubic computational cost.

6.3 Task 2: High-Curvature Landscapes (AUTOENCODER)

We utilize a Deep Autoencoder on FashionMNIST. This task tests robustness in ill-conditioned landscapes.

Results. D-Muon achieves the lowest reconstruction error (0.011), confirming that exact orthogonalization is highly effective for deep bottleneck architectures. However, ANDI ($\eta = 0.09$) demonstrates superior robustness compared to diagonal methods, achieving a validation loss of 0.015 versus AdamW's 0.019. While D-Muon wins on pure performance, ANDI bridges the gap between diagonal and spectral methods, offering significantly better convergence than AdamW with lower computational cost than Muon.

6.4 Task 3: Convolutional Generalization (RESNET)

We train a ResNet-9 on CIFAR-10 to evaluate performance on convolutional topologies.

Results. D-Muon dominates this task (Val Loss 0.277), followed by AdamW (0.338). ANDI trails closely with a loss of 0.358.

Table 2: **Benchmarking Summary.** Best validation loss (lower is better) and total wall-clock time. ANDI achieves SOTA performance on Transformers (GPT) with minimal overhead compared to AdamW.

	GPT		Autoencoder		ResNet	
Optimizer	**Val Loss**	**Time (s)**	**Val Loss**	**Time (s)**	**Val Loss**	**Time (s)**
AdamW	1.577	**103**	0.0199	**72**	0.338	**245**
D-Muon	1.555	127	**0.0112**	81	**0.277**	321
ANDI (Ours)	**1.535**	108	0.0156	74	0.358	255

6.5 Task 4: Instruction Fine-Tuning (LLAMA)

We fine-tune a quantized Llama-3 8B model [3] on the Alpaca-GPT4 dataset using Low-Rank Adaptation (LoRA) [6] ($r = 16, \alpha = 16$). This setting explicitly tests optimization in constrained, rectangular subspaces ($\Delta W \in \mathbb{R}^{d\times r}$).

Results. Figure 2 demonstrates the "Fast-Start" phenomenon of ANDI.

- **Convergence Speed:** ANDI (η = 5e-2) rapidly reduces loss, achieving a validation loss of ~ 1.6 within 60 steps. AdamW (η = 1e-3) stagnates early, remaining above 2.0 loss for the first 100 steps.
- **Subspace Equilibration:** While D-Muon ($\eta = 0.02$) performs well, it converges slower than ANDI in the initial phase. We hypothesize that Muon's strict orthogonality constraint is overly restrictive for the rectangular geometry of LoRA adapters (4096×16). ANDI's additive normalization naturally balances the disparate scales of the high-dimension input and low-dimension rank without forcing rotational invariance, leading to faster adaptation.

7 Discussion

The experimental results presented in Section 6 highlight distinct behaviors of ANDI relative to the baselines. Here, we interpret these findings through the lens of our theoretical framework.

7.1 Inductive Bias in Transformers

The most significant result is the superior performance of ANDI on the GPT task (Figure 1, Left). We hypothesize this is due to the specific structure of Transformer weights. In Self-Attention, rows and columns correspond to specific embedding features or heads. If a specific feature dimension has high variance, it dominates the LayerNorm statistics. AdamW exacerbates this by normalizing based on historical variance, potentially stalling the learning of that feature. ANDI, utilizing instantaneous spatial norms, enforces an equilibrium where all feature dimensions contribute more equally to the update step. This aligns well with the "feature competition" dynamic often observed in Transformer training [15].

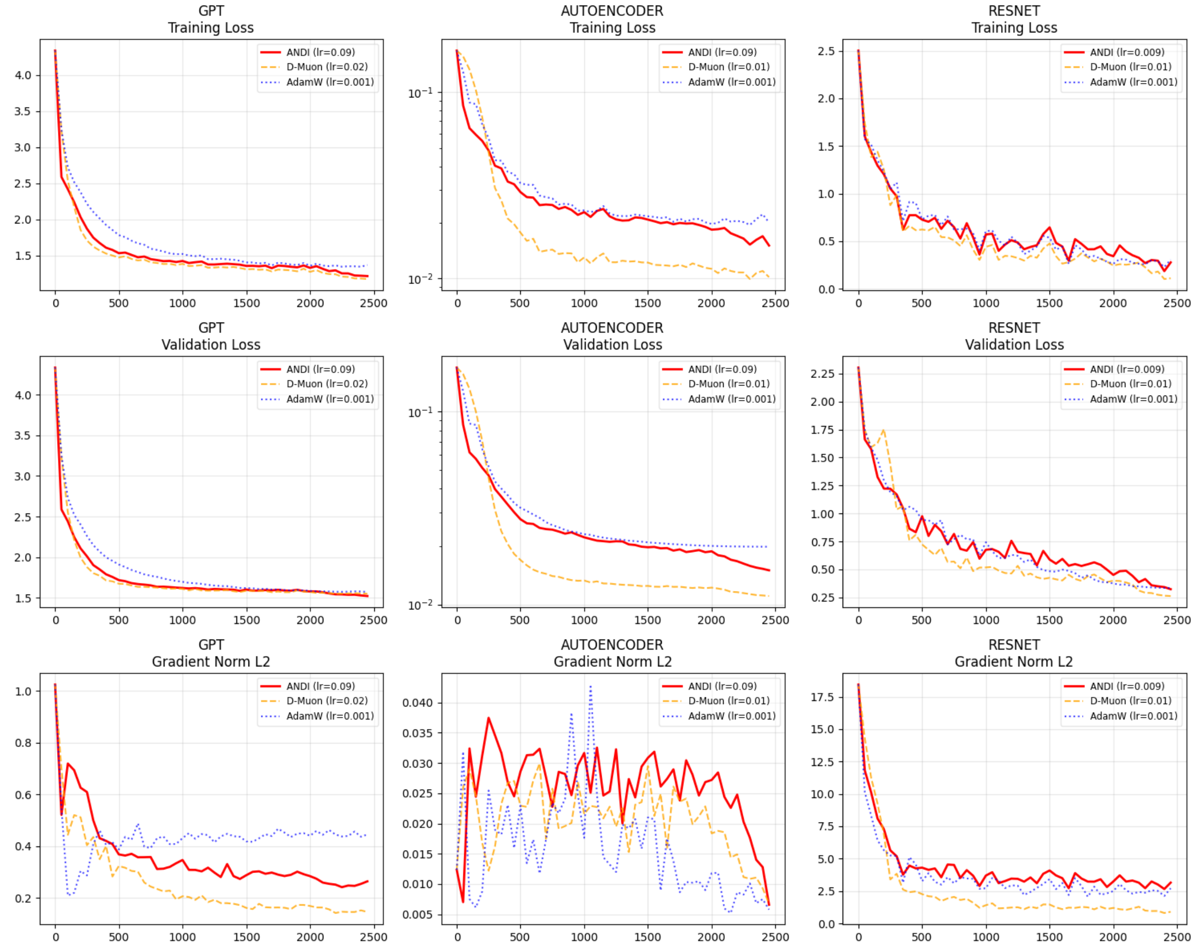

Figure 1: **Optimization Trajectories (Tasks 1-3).** Training loss, Validation loss, and Gradient Norm evolution. ANDI (Red) demonstrates superior convergence on GPT, outperforming both D-Muon (Orange) and AdamW (Blue). On Autoencoder, ANDI significantly outperforms AdamW, bridging the gap to the computationally expensive D-Muon.

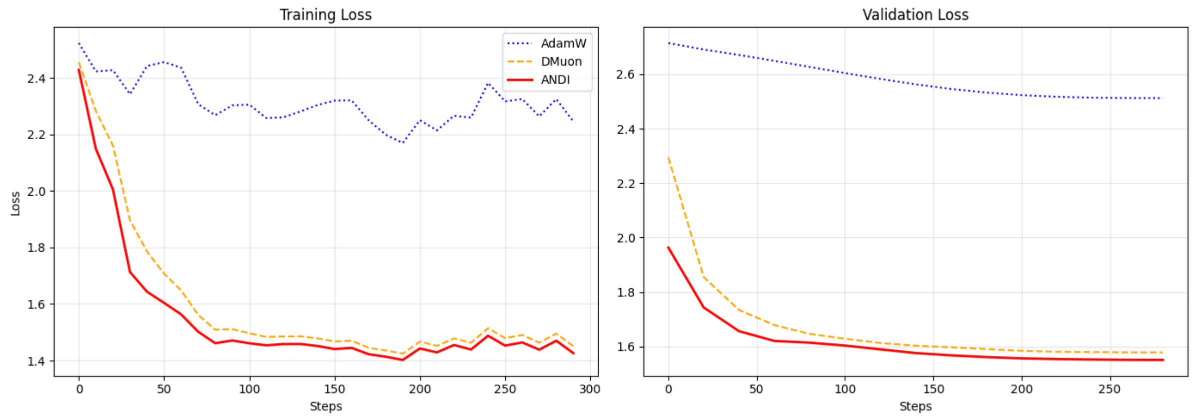

Figure 2: **Llama-3 8B Fine-Tuning (LoRA).** Training (Left) and Validation (Right) loss. ANDI (Red) dominates the optimization trajectory, exhibiting a "fast-start" behavior that significantly outpaces AdamW (Blue) and converges faster than D-Muon (Orange) in the low-rank regime.

ANDI slightly outperforms D-Muon on the GPT pre-training task (Figure 1, Left) and Llama-3 8B fine-tuning (LoRA) task (Figure 2). D-Muon operates by orthogonalizing the gradient update, which effectively equilibrates the spectrum of the weight update. ANDI achieves a similar effect through row-column normalization. By dividing G_{ij} by $\|\mathbf{g}_{i,:}\|+\|\mathbf{g}_{:,j}\|$, ANDI penalizes updates in directions where the aggregate gradient mass is high. This creates a "pseudo-whitening" effect. The first-order approximation provided by ANDI captures the majority of the structural benefit at a fraction of the compute cost.

7.2 Stability vs. Aggressiveness

In the AUTOENCODER task, Muon exhibited rapid initial descent followed by stagnation. Spectral methods are known to be "aggressive"—they project gradients onto the Stiefel manifold, which can be suboptimal in landscapes requiring fine-grained magnitude adjustments (like MSE loss). ANDI's energy-consistent scaling (Equation 4) allowed it to interpolate between the structural regime (early training) and the magnitude-sensitive regime (fine-tuning), preventing the premature plateauing observed in D-Muon.

7.3 Limitations

While ANDI achieves spectral efficiency, the current formulation entails specific structural constraints and approximations:

1. **Regime Heterogeneity:** The proposed equilibration operator relies on the existence of orthogonal tensor modes to estimate local curvature. For parameters with dimension $d < 2$ (biases, gains) or matrices falling below the threshold δ, ANDI reverts to global energy normalization. In this regime, the algorithm approximates the dynamics of *Normalized Gradient Descent*, foregoing the element-wise equilibration benefits. Consequently, the structural advantages of ANDI are primarily realized in "wide" layers (Linear, Conv2d, Attention), while vector parameters do not benefit from coordinate-aware preconditioning.
2. **Spatial Homogenization via Matricization:** For higher-order tensors, such as convolutional kernels $\mathcal{K} \in \mathbb{R}^{C_{out}\times C_{in}\times H\times W}$, ANDI applies a matricization step to reshape the tensor into $\mathbb{R}^{C_{out}\times(C_{in}HW)}$. While this efficiently captures correlations between output filters, it treats spatial dimensions and input channels uniformly. This approximation ignores the local inductive bias of the convolution operator, potentially under-utilizing the specific spatial geometry of the kernel.
3. **Discrete Structural Transition:** A minimal dimensionality δ is required to compute statistically meaningful marginal norms. This imposes a discrete decision boundary where the optimizer switches between coordinate-wise equilibration (for large matrices) and global spherical constraints (for small tensors).

7.4 Broader Theoretical Implications: Additive Equilibration as a Primitive

While derived within the context of non-convex optimization, the core operator of ANDI—*Rank-1 Additive Equilibration*—possesses algebraic properties that render it a tractable primitive for domains requiring structural regularization. We define the general operator $\Phi(\mathbf{X})$ on a matrix $\mathbf{X} \in \mathbb{R}^{m\times n}$ with row norms $\mathbf{r}$ and column norms $\mathbf{c}$ as:

$$\Phi(\mathbf{X})_{ij} = \frac{X_{ij}}{r_i + c_j + \epsilon}. \tag{22}$$

We identify three regimes where this transformation offers a geometric advantage over standard baselines.

Batch-wise Density Normalization in Retrieval. In dense retrieval, the *Hubness Phenomenon* [10] degrades performance when high-density vectors ("hubs") appear as nearest neighbors to a disproportionately large number of queries. Standard cosine similarity normalization fails to account for global density variations. Consider a batch similarity matrix $\mathbf{S} \in \mathbb{R}^{B\times N}$ where $B > 1$. The column norm $\|S_{:,j}\|_2$ serves as a local proxy for the density of the j-th database item relative to the query batch. Applying ANDI yields a re-scored matrix $\widehat{\mathbf{S}}$:

$$\widehat{S}_{ij} = \frac{S_{ij}}{\|S_{i,:}\|_2 + \|S_{:,j}\|_2}. \tag{23}$$

By explicitly incorporating the column energy, ANDI penalizes "hub" items that possess high aggregate similarity across the batch. Unlike CSLS [1], which requires expensive k-NN search for density estimation, ANDI achieves structural suppression in $\mathcal{O}(BN)$ (linear in the similarity matrix size), efficiently sharpening the retrieval distribution around semantically unique correspondences.

Soft Redundancy Suppression in Representation Learning. Standard objectives for feature decorrelation (e.g., in self-supervised learning) often rely on Pearson correlation, which normalizes covariance entries via the geometric mean of variances: $\rho_{ij} = \mathcal{C}_{ij}/\sqrt{\mathcal{C}_{ii}\mathcal{C}_{jj}}$. ANDI implies a stricter regularization scheme. By the AM-GM inequality, the arithmetic mean of structural norms dominates the geometric mean: $r_i + c_j \geq 2\sqrt{r_i c_j}$. When applied to a feature covariance matrix $\mathcal{C}$, ANDI acts as a *Soft Redundancy Suppression* operator. Because the row norm $r_i = \|\mathcal{C}_{i,:}\|_2$ captures the total correlation energy of feature i with all other dimensions (not just its own variance), ANDI naturally down-weights features that are redundant (i.e., highly correlated with the global set). This serves as an $\mathcal{O}(N^2)$ proxy for feature selection, discouraging representational collapse without the computational cost of whitening (ZCA).

One-Shot Structural Balancing for Graph Matching. The Sinkhorn-Knopp algorithm [2] solves entropic optimal transport by iteratively projecting a matrix onto the Birkhoff polytope (doubly stochastic matrices). ANDI functions as a closed-form, one-shot approximation of this balancing objective. While it does not enforce strict mass conservation ($\sum P = 1$), it enforces element-wise boundedness (as proven in Proposition 2). In Deep Graph Matching, where the goal is robust node correspondence, the raw similarity matrix is often dominated by outliers. ANDI equilibrates the energy distribution of the matrix, mapping values to a bounded range while penalizing nodes with disproportionately high matching degrees. This provides a conditioned initialization for matching layers, capturing the structural benefits of Sinkhorn scaling without the latency of iterative solvers.

8 Conclusion

We presented ANDI, an algorithm designed to democratize efficient optimization in deep learning. By replacing expensive spectral decompositions with row-column additive normalization, ANDI successfully navigates the trade-off between computational efficiency and geometric awareness.

Our theoretical analysis confirmed that ANDI preserves descent directions and bounds update magnitudes, ensuring robust stability. Empirically, ANDI demonstrated state-of-the-art convergence rates on non-convex vision and language modeling benchmarks, proving particularly effective for Transformers. These results suggest that the heavy machinery of second-order or spectral optimization is not always necessary; simple, algebraically sound equilibration of gradient matrices can yield equivalent gains with the efficiency of standard first-order methods. Future work will investigate the application of ANDI to distributed training setups, where its low memory footprint and lack of collective communication overheads could offer significant wall-clock speedups.

A Experimental Methodology and Implementation Details

To ensure a rigorous evaluation of the ANDI optimizer, we aligned our experimental protocol with the comprehensive benchmarking standards recently established by Semenov et al. [12]. Their work highlights critical flaws in common optimizer comparisons—specifically regarding weight decay application in spectral methods and the coupling of learning rate schedules to specific optimizer families. Below, we detail how these recommendations were translated into our implementation.

A.1 Optimizer Selection: Dispersed Muon (D-Muon)

For the spectral baseline, we utilize the **Dispersed Muon (D-Muon)** algorithm rather than the original Muon implementation.

- **Rationale:** As demonstrated in Section A.3 and Figures 4 & 8 of [12], the original Muon optimizer suffers in extended training regimes because it fails to apply weight decay to 2D matrix parameters. D-Muon corrects this by applying decoupled weight decay to all parameter groups prior to the update step.
- **Implementation:** Our code applies Newton-Schulz orthogonalization only to tensors with dimensions $> 32 \times 32$ (Linear and Conv2d layers), falling back to an AdamW-style backend for vector parameters (biases, normalization scales). This hybrid approach is consistent with the "Dispersed" methodology required to maintain stability (Takeaway 3 in [12]).

- **Newton-Schulz Iterations:** We fix the number of orthogonalization steps at $N_{NS} = 5$. Semenov et al. (Figure 37) show that increasing iterations beyond 5 yields diminishing returns in convergence while incurring unnecessary wall-clock overhead.
- **Momentum:** We utilize momentum with $\beta = 0.95$. Extensive hyperparameter tuning on Llama architectures (Tables 40-41 in [12]) identifies 0.95 as consistently superior to the standard 0.9 or 0.99 for Muon-based methods.

A.2 Learning Rate Schedulers

We strictly adhere to the finding that different optimizer classes require distinct learning rate schedules to achieve optimal performance (Takeaway 6 in [12]).

- **Coordinate-wise Methods (AdamW):** We utilize a **Cosine Decay** schedule. Benchmarks (Figure 11c [12]) show that AdamW consistently outperforms other schedules (linear or WSD) under cosine decay.
- **Spectral Methods (D-Muon) and ANDI:** We utilize a **Warmup-Stable-Decay (WSD)** schedule. Semenov et al. demonstrate that spectral methods exhibit a unique preference for WSD (Figure 11a), often plateauing early with standard cosine annealing. Consequently, using Cosine for AdamW and WSD for ANDI/D-Muon represents the fairest comparison, allowing each optimizer to operate under its ideal conditions.

A.3 Evaluation Metrics

Validation Loss vs. Training Loss. Following the reporting standards urged in [12] (Figure 1 & Takeaway 12), we report **Final Validation Loss** as the primary metric. Training loss is frequently misleading in optimizer benchmarks due to varying degrees of overfitting and regularization handling (e.g., weight decay impacts training loss values differently across optimizers).

Wall-Clock Time. Step count is an insufficient metric for spectral methods due to the computational overhead of matrix operations. As noted in Appendix D.3 and Figure 18 of [12], methods like SOAP or Muon can suffer from slowdowns as model size increases. We report total wall-clock time to quantify the trade-off between convergence speed (per step) and computational throughput.

A.4 Hyperparameter Search and Configuration

While we adopt the architectural hyperparameters suggested by the references, we performed an independent robust search for learning rates to accommodate the specific scale of our proxy tasks.

- **Search Space:** We performed a logarithmic grid search for learning rates: $\eta \in \{1\text{e-}3, \ldots, 2\text{e-}4\}$ for AdamW and $\eta \in \{0.09, \ldots, 0.009\}$ for ANDI and D-Muon.
- **Weight Decay:** Fixed at $\lambda = 0.01$ for all methods. While Semenov et al. explore values up to 0.1, we selected 0.01 to ensure consistent regularization strength across the diverse architectures (ResNet vs. Transformer) in our suite.
- **Gradient Clipping:**
  - *Tasks 1-3 (Proxy Benchmarks):* We monitored gradient norms but did not enforce clipping (infinite threshold) to observe the raw optimization trajectory and stability of the updates.
  - *Task 4 (LLAMA Fine-tuning):* We applied standard gradient clipping (norm 1.0) via the Unsloth framework. This aligns with findings (Takeaway 17 in [12]) that clipping is crucial for stability in lower-precision (4-bit/bfloat16) regimes involving large rectangular matrices.

A.5 Hardware and Environment

All benchmarks were executed on a single NVIDIA Tesla T4 (16GB) available in the free tier of Google Colab. The Llama-3 fine-tuning was performed using 4-bit quantization via the Unsloth framework to fit within memory constraints. Seeds 42 and 1337 were used for reproducibility.

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